How to Eat Mushrooms in the Wild Without Getting Killed: Active Learning for Fungal Edibility Classification

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1. Background

1.1. Fine-Grained Fungal Image Classification

Eating wild mushrooms is a popular activity and a small industry in many parts of the world, but every year thousands of people end up in the hospital because they mistook a poisonous species for an edible one. There are more than 14,000 known mushroom species, and many of them look almost the same even though one is safe and the other is not. The most famous example is the Amanita phalloides (Death Cap), which can kill a person but looks similar to several edible Agaricus mushrooms. This issues make mushroom classification a good test case for machine learning methods that care about safety and not only accuracy. The mushrooms look alike, calling a toxic one "safe" is much worse than the other way around, and some species have far more photos than others.

The Danish Fungi 2020 (DF20) benchmark from Picek et al. [1] is one of the best public datasets we found for this task. The images come from the Atlas of Danish Fungi, where people submit photos of mushrooms they have seen. DF20 has a few nice properties for our work. The species labels are checked against the official taxonomy, each photo comes with extra information like location and habitat. The last point matters because it means we can use ImageNet-pretrained models without worrying that the model has already seen the test images. In their paper, Picek et al. trained a Vision Transformer (ViT-B/16-224) on DF20 and got about 0.72 top-1 species accuracy and 0.74 macro F1. The same group later released a smaller version called DF20-Mini, which keeps only the species that have enough images to train on. This makes DF20-Mini more suitable for machine learning, since every class has a reasonable number of training examples instead of just one or two, which can be used as a backbone for other tasks.

A related line of work by Picek et al. [2] used a similar model in the FungiVision citizen-science app, which shows that large-scale mushroom recognition can actually work in practice when there is enough

labeled data to train on. Our project looks at the opposite situation, that is, what can we do when labeled data is scarce.

1.2. Active Learning Speeds Up the Labeling Process

Active learning (AL) is a way of training a model where, instead of labeling all the data at once, the model picks which examples it wants labeled next. The idea is to reach a good accuracy with fewer total labels than you would need if you just picked images at random [3]. Suppose we have a labeled set L, an unlabeled pool U, a model with parameters θ, and we are allowed to label B new examples each round. An AL strategy is a scoring function $a(\cdot)$ that picks the best batch of B examples from the unlabeled pool,

$$Q = \underset{Q \subseteq U,\ |Q|=B}{\arg\max}\ a(Q;\theta, L, U).$$

After labeling these examples, we add them to the labeled set, $L \leftarrow L \cup Q$, and retrain the model. This loop repeats until the labeling budget runs out.

There are several common families of AL strategies, and each one has a different idea about what makes an example worth labeling. Uncertainty sampling picks examples that the current model is least sure about. Diversity sampling picks examples that look different from what is already in the labeled set, so the labeled data covers more of the feature space [4]. Bayesian methods like BALD [5] try to find the examples that would teach the model the most, by separating the part of the uncertainty that comes from the model itself from the part that comes from noise in the data. Class-balanced methods add a weighting term that pushes the model to ask for more examples of rare classes, which helps a lot on long-tailed datasets [7].

1.3. From Species Prediction to Edibility Prediction

When someone is out in the woods picking mushrooms, they do not really need to know the exact species. What they need to know is whether the mushroom is safe to eat. So instead of stopping at species-level prediction, a four-way edibility decision is prefered. The four buckets are SAFE (edible raw or cooked), COOK_FIRST (only safe after proper cooking), AVOID (toxic, deadly, or just inedible), and UNCERTAIN (the literature does not agree, or there is not enough information).

It is important to note that the four buckets are not the training target. The model is still trained on all 179 species, and we only convert the species prediction into a bucket prediction at the end. Let g be the

function that maps each species c to its bucket k, so $g : \{1,\dots,C\} \to \{1,\dots,K\}$. There are two natural ways to do this conversion. The first is hard decoding, where we first take the most likely species and then look up its bucket,

$$\hat{b}_{\text{hard}}(x) \;=\; g\left(\arg\max_{c}\; p(y=c \mid x)\right). \tag{1}$$

The second is soft decoding, where we add up the predicted probabilities of all species inside each bucket and pick the bucket with the largest total,

$$\hat{b}_{\text{soft}}(x) \;=\; \arg\max_{k} \sum_{c:\, g(c)=k} p(y=c \mid x). \tag{2}$$

The nice thing about Equation (2) is that it forgives mistakes between species that share a bucket. For example, if the model is confused between two edible *Boletus* species, the bucket prediction can still come out right. We test whether this actually helps in our experiments.

2. Significance

Most fine-grained image classification work focuses on getting the highest top-1 accuracy and treats every error the same way. But in a real foraging setting, the cost of errors is not even. Mistaking a Death Cap for an edible Agaricus can kill people, while mixing up two edible Russula species does not really matter. We handle this in two ways. First, instead of stopping at the species prediction, we group the 179 species into four edibility buckets and check how well the model does on those buckets. Second, on top of the usual accuracy numbers, we also report the recall for each bucket, so we can see how often the model misses the toxic ones. This way, the evaluation matches what a real user would care about.

The active learning literature has lots of strategies, but it is not very clear which one to use for a fine-grained, long-tailed, safety-critical task like ours. We compare six strategies across five seeds and report six different metrics, including bucket accuracy, macro F1, and per-bucket recall. As far as we know, this is the first benchmark of this kind on DF20-Mini. One thing we found is that the ranking of strategies changes a lot depending on the metric. For example, BALD could be the best on overall bucket accuracy, but class-balanced entropy may be much better at catching the UNCERTAIN class. This kind of trade-off matters for any AL deployment where the metric the user cares about is not the same as the loss the model was trained

on.

Annotation is also the bottleneck for extending mushroom recognition beyond Denmark. Labeling fungal images requires expert mycologists, and they are hard to find. So cutting the labeling cost is not just a nice engineering result, it is what makes it realistic to build similar systems for new regions or for rare species, where the safety benefit is the largest. To show how much labeling effort active learning can save, we compare each strategy against a full-supervision ceiling, where the model is trained on the entire labeled pool. This suggests that a usable safety-aware classifier for a new region could be built with a manageable annotation budget.

3. Specific Aims

The overall goal of this project is to find out whether modern active learning strategies can build a useful, safety-aware mushroom classifier without needing too many labels, by mapping images into the four edibility buckets defined earlier. We further break this down into three smaller aims.

Aim 1. Build an edibility label hierarchy on top of the species labels. We use a large language model (Claude Opus 4.7) to assign each of the 179 DF20-Mini species to one of four edibility buckets. Then we check whether the resulting hierarchy makes sense by looking at how the buckets are split across the six genera, how balanced the buckets are at the image level, what kinds of toxicity end up inside the AVOID bucket, and how confident the LLM was for each bucket. The output is an enriched label table that supports both species-level training and bucket-level evaluation through Equations (1) and (2).

Aim 2. Build an active learning pipeline using embeddings data. We extract 768-dimensional embeddings from the ViT-B/16-224 backbone released by Picek et al. [1], which was already fine-tuned on DF20-Mini. On top of these embeddings, we train a small MLP head with MC-dropout, and run a low-budget AL loop with $N_{\text{init}} = 400$ initial labels, a batch size of $B = 100$ per round, and a total budget of 5,000 labels. We then compare six query strategies (random, entropy, margin, core-set, class-balanced entropy, and BALD) using five random seeds, and we also train a fully supervised model on the entire pool to get an upper bound.

Aim 3. Evaluate the strategies. We report species accuracy, species macro F1, bucket accuracy under both hard and soft decoding, per-bucket recall, and a Pareto plot of overall accuracy against UNCERTAIN recall. We also measure label efficiency by counting how many labels each strategy needs to reach 80%, 90%, and 95% of the full-supervision ceiling. Finally, we run a per-genus analysis with extra attention to the genus Amanita, which contains the most dangerous species in the dataset.

4. Experimental Design

4.1. Dataset and Label Hierarchy

We use DF20-Mini [1] with its official train and test split. After dropping the rows with missing species labels, the training pool has about 32,724 images and the test set has about 4,000 images. Together they cover $C = 179$ species across 6 genera (*Amanita*, *Boletus*, *Russula*, *Mycena*, *Agaricus*, and *Clitocybe*).

To get the edibility labels, we ask Claude Opus 4.7 about each species and have it return both an edibility label and a confidence rating in $\{\text{high}, \text{medium}, \text{low}\}$. We then map the LLM output into one of the $K = 4$ buckets using the rule

$$g(\text{species}) = \begin{cases} \text{SAFE} & \text{if the LLM label is edible,} \\ \text{COOK_FIRST} & \text{if the LLM label is edible_cooked_only,} \\ \text{AVOID} & \text{if the LLM label is one of } \{\text{poisonous, deadly, inedible}\}, \\ \text{UNCERTAIN} & \text{if the LLM label is unknown.} \end{cases}$$

At the image level, this gives us roughly 12% SAFE, 20% COOK_FIRST, 55% AVOID, and 13% UNCERTAIN, which is much more balanced than the 179-way species distribution where the most common species has about 39 times more images than the rarest one.

4.2. Backbone and Embedding Extraction

To keep our experiments focused on the AL strategy rather than on representation learning, we use a frozen feature extractor. We take the public DF20M ViT-B/16-224 checkpoint from Picek et al. [1] and remove the classification head, then run every DF20-Mini image through the backbone to get a $d = 768$-dimensional

embedding,

$$\phi : \mathbb{R}^{3\times224\times224} \to \mathbb{R}^{768}, \qquad \tilde{\phi}(x) = \phi(x)/\|\phi(x)\|_2.$$

We ℓ_2-normalize the embeddings so that cosine similarity is well-defined, which is needed by the core-set strategy. As a sanity check, fitting a logistic regression on $\tilde{\phi}(x)$ gives roughly the same species accuracy that Picek et al. report, which tells us the checkpoint loaded correctly.

4.3. The MC-Dropout MLP Classifier

The classifier head $f_\theta : \mathbb{R}^{768} \to \Delta^{C-1}$ is a small MLP,

$$f_\theta = \text{Linear}_{768\to512} \to \text{ReLU} \to \text{Drop}_p \to \text{Linear}_{512\to256} \to \text{ReLU} \to \text{Drop}_p \to \text{Linear}_{256\to179},$$

with dropout probability $p = 0.4$. Following Gal and Ghahramani [6], we keep dropout turned on at inference time, so $T = 20$ stochastic forward passes give us T slightly different predictions $\{p(y \mid x, \theta_t)\}_{t=1}^{T}$ that act like samples from an approximate posterior. These are what BALD needs.

First, we retrain the head from scratch every AL round instead of fine-tuning it from the previous round. We tried the warm-start version and saw that the BALD scores got biased toward the dropout masks that happened to be active in the previous training, so retraining from scratch is the cleaner option. Second, we use two hidden layers with $p = 0.4$. With only one hidden layer, the model becomes too confident on the small early labeled sets and BALD stops working properly. Training uses AdamW with learning rate 10^{-3}, weight decay $\lambda = 5 \times 10^{-4}$, batch size 128, and 30 epochs, with the cross-entropy loss

$$\mathcal{L}(\theta) = -\frac{1}{|L|} \sum_{(x,y)\in L} \log p(y \mid x, \theta) + \tfrac{\lambda}{2}\|\theta\|_2^2.$$

We picked these hyperparameters by checking train and test accuracy every five epochs to see when the model converged.

4.4. Active Learning Loop

We use a low-budget AL setup that mimics a realistic labeling project where expert time is the bottleneck.

Initial labeled set. We start with $N_{\text{init}} = 400$ labeled images, drawn by stratified sampling so that each of the 179 species gets about 2 examples. This avoids the cold-start problem where some classes are never

seen by the model in the first few rounds.

Batch size and total budget. At each round, the strategy picks $B = 100$ new images to label. The loop stops once the labeled set reaches 5,000 images, which is about 15% of the full training pool.

Per-round procedure. In each round, we retrain the MLP head from scratch on the current labeled set, evaluate it on the held-out test set, and use the trained model to score the unlabeled pool. The top $B = 100$ images are then added to the labeled set for the next round.

Seeds and reporting. We repeat each strategy with five random seeds and report the mean learning curve, with the ± 1 standard deviation across seeds shown as a shaded band. As an upper bound, we also train the same MLP head on the entire training pool to get a full-supervision ceiling.

4.5. Query Strategies

We compare six strategies in total, covering the four families described in the Background.

Random. The passive baseline. Each round it samples B examples uniformly from the unlabeled pool, $Q_r \sim \text{Uniform}(U_r)$.

Entropy. Picks the top B examples by predictive entropy,

$$H[p(y \mid x)] \;=\; -\sum_{c=1}^{C} p(y=c \mid x) \log p(y=c \mid x).$$

Entropy is high when the model spreads its probability across many classes, so this strategy goes after the examples the model is least sure about.

Margin. Picks the bottom B examples by the gap between the top two predicted probabilities,

$$m(x) \;=\; p_{(1)}(x) - p_{(2)}(x).$$

A small $m(x)$ means the model is torn between two classes, so this strategy targets borderline cases.

Core-set [4]. A diversity-based method. It picks the example that is farthest from anything we have already labeled, using one minus cosine similarity in the embedding space,

$$x^* \;=\; \arg\max_{x \in U} \min_{x' \in L} d(\tilde{\phi}(x), \tilde{\phi}(x')), \qquad d(x,x') = 1 - \tilde{\phi}(x)^\top \tilde{\phi}(x').$$

We pick B samples greedily, updating the minimum distance after each selection.

Class-balanced entropy [7]. A long-tail-aware version of entropy. We multiply the entropy score by a class-frequency weight that is bigger for classes we have seen less,

$$w_c \;=\; \frac{1}{\sqrt{n_c+1}}, \qquad x^* \;=\; \arg\max_{x\in U}\; H[p(y\mid x)]\cdot w_{\hat{y}(x)},$$

where n_c is how many examples of class c are already in the labeled set and $\hat{y}(x)$ is the model's predicted class for x.

BALD [5]. A Bayesian strategy. We approximate the BALD score with $T = 20$ MC-dropout forward passes,

$$\widehat{\mathrm{BALD}}(x) \;=\; H\left[\tfrac{1}{T}\sum_{t=1}^{T} p(y\mid x,\theta_t)\right] \;-\; \tfrac{1}{T}\sum_{t=1}^{T} H[p(y\mid x,\theta_t)]\,,$$

which approximates the part of the model's uncertainty that comes from the model itself rather than from noise in the data. The strategy picks the top B samples by this score.

For comparison, we also train the same MLP on the entire labeled pool ($L = U$) to get a full-supervision ceiling.

4.6. Evaluation Metrics

For each (strategy, seed, round) triple, we record six headline numbers. These are species top-1 accuracy, species macro F1, bucket accuracy under both hard and soft decoding (Equations (1) and (2)), bucket macro F1 under both schemes, and per-bucket recall,

$$\mathrm{Recall}_k \;=\; \frac{|\{i : \hat{b}(x_i) = k,\; b(x_i) = k\}|}{|\{i : b(x_i) = k\}|}.$$

We also store per-genus species accuracy, so we can later check how much of the headline performance comes from each genus.

Label efficiency. For each target $\tau \in \{0.80, 0.90, 0.95\}$ and each strategy, we find the smallest budget at which the strategy crosses τ times the full-supervision accuracy,

$$N_\tau(\text{strategy}) \;=\; \min\bigl\{n : \mathrm{acc}_{\text{strategy}}(n) \geq \tau\cdot \mathrm{acc}_{\text{full-sup}}\bigr\}.$$

If a strategy fails to reach the target inside the 5,000-label budget for a particular seed, that seed is left out of the mean and we report the success rate (for example $4/5$) alongside the number.

4.7. Safety-Focused Analysis

On top of the headline numbers, we run three extra analyses that focus on the safety side of the problem. First, we plot AVOID recall and UNCERTAIN recall as their own learning curves. The AVOID curves end up tightly clustered, which means toxic species are caught reliably no matter which strategy we use. The UNCERTAIN curves are much more spread out, which suggests that the choice of strategy matters most when the goal is flagging ambiguous specimens. Second, we make a Pareto plot of bucket accuracy against UNCERTAIN recall at the final budget. BALD is the best on overall accuracy but drops to fifth on UNCERTAIN recall, while class-balanced entropy goes the other way. Third, we build a per-genus heatmap of bucket recall. The lethal genus *Amanita* reaches an AVOID recall of 0.89 even when the species accuracy is only around 0.65, which shows the benefit of the soft-aggregation decoder in Equation (2).

5. Results

We present the results in the same order as the three specific aims: the LLM-derived edibility hierarchy (Aim 1), the embedding-quality sanity check and the full-supervision ceiling (Aim 2), and the head-to-head comparison of the six AL strategies (Aim 3). All AL numbers are reported as the mean across five seeds, with ± 1 standard deviation in parentheses where space allows.

5.1. The Edibility Hierarchy is Balanced and Epistemically Coherent

Mapping the 179 DF20-Mini species through the LLM and the rule in Section 4.1 yields 21 SAFE species (11.7%), 37 COOK_FIRST (20.7%), 96 AVOID (53.6%), and 25 UNCERTAIN (14.0%). At the image level the picture flattens to 15.0% / 21.0% / 57.0% / 7.0% across the 36,361 images (Figure 1). Compared with the raw 179-way distribution, where the most populated species has roughly $39\times$ the support of the rarest, the four-bucket view is far more tractable for classification while still preserving the safety semantics that motivate the project.

The per-genus composition (Figure 2) shows that the six genera present qualitatively different sub-problems. *Mycena* is uniformly AVOID and is therefore trivial at the bucket level even when species-level

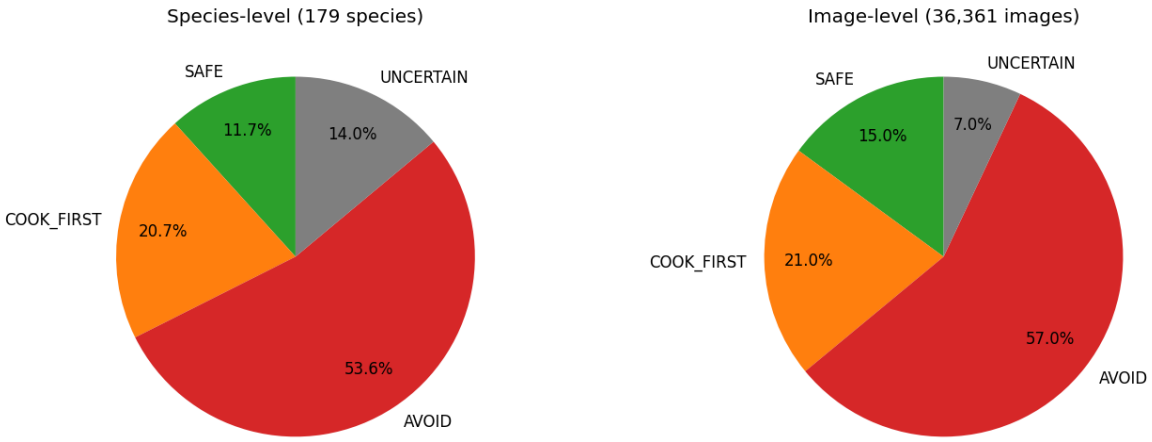


Figure 1: Bucket distribution at the species level (left) and the image level (right). The image-level view is what the classifier actually sees and is substantially more balanced than the raw species distribution.

accuracy is poor. *Boletus* is split only between SAFE and COOK_FIRST, again easy at the bucket level. The remaining four genera (*Agaricus*, *Amanita*, *Clitocybe*, *Russula*) are 3- or 4-way mixtures, and *Russula* alone contributes 16 of the 25 UNCERTAIN species. The *Amanita* entry is the most safety-critical: 9 of its 17 species map to AVOID and include the deadliest taxa in the dataset.

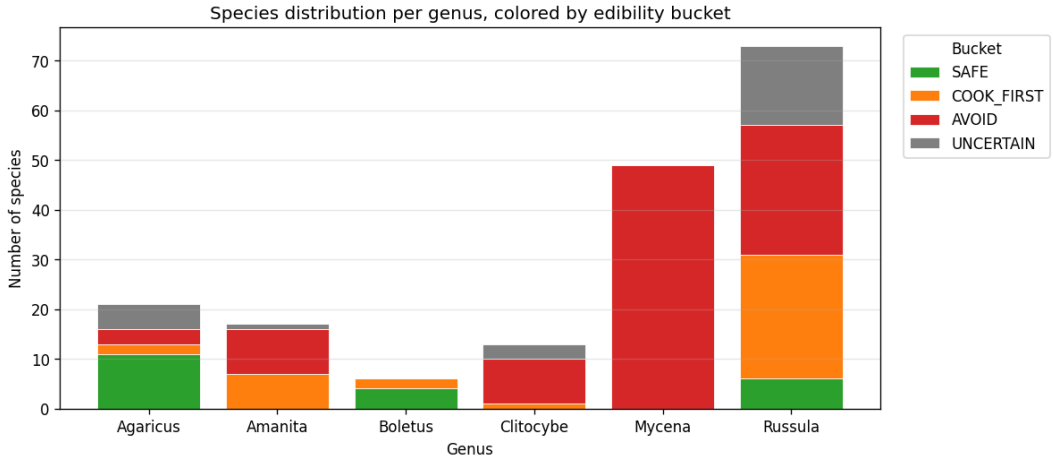


Figure 2: Number of species per genus, coloured by edibility bucket. *Mycena* is uniformly AVOID; *Russula* is the dominant contributor to UNCERTAIN; *Amanita* concentrates AVOID risk.

The internal composition of the AVOID bucket is heterogeneous: 52 species are flagged *poisonous*, 42 are *inedible*, and only 2 are *deadly*; on the LLM toxicity scale the counts are 43 at level 1, 46 at level 2, 5 at level 3, and 2 at level 5. AVOID therefore combines emergency-level danger with mere unpalatability and could be split further if a future deployment wants differential warnings.

The LLM confidence distribution (Figure 3) validates UNCERTAIN as a genuine epistemic category rather than a dumping ground: 96% of UNCERTAIN species came back with the LLM's *low* confidence flag, while 81% of SAFE species came back as *high* and no SAFE species was rated *low*. The agreement between the bucket and the confidence label is what we wanted before treating UNCERTAIN as a research signal.

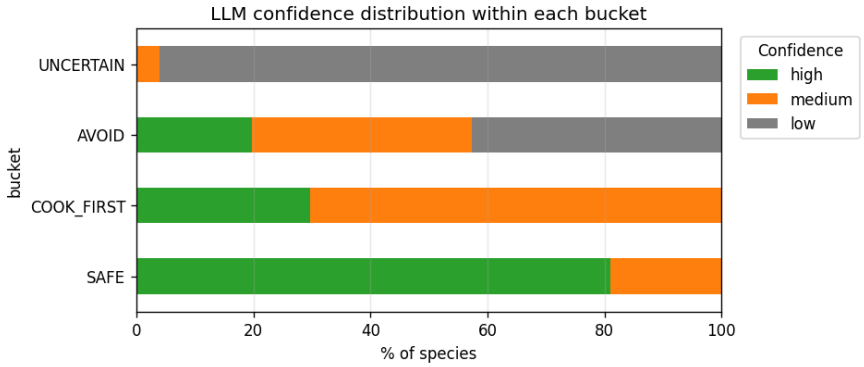


Figure 3: LLM confidence within each bucket. UNCERTAIN is dominated by *low* confidence and SAFE by *high*, supporting the use of UNCERTAIN as a distinct epistemic category.

5.2. Embedding Sanity Check and Full-Supervision Ceiling

Loading the public DF20M ViT-B/16-224 checkpoint and extracting ℓ_2-normalized 768-dimensional features takes about 20 minutes for the full 36,361 images on Apple Silicon (MPS). A logistic regression fit on these features reaches 0.981 train and **0.626** test species accuracy, in the 0.60–0.72 range we expected from the original backbone [1]. This rules out a checkpoint-loading bug and confirms that the embedding space already carries most of the species signal.

Training the MC-dropout MLP head on the entire labeled pool (32,724 images) gives the full-supervision ceiling reported in Table 1. Bucket accuracy is high in absolute terms but the per-bucket recall is very uneven: AVOID is recovered at 0.895 while UNCERTAIN sits at only 0.535. This 0.36 gap is the headroom that strategy choice will be measured against in Section 5.4. Soft decoding (Equation 2) lifts overall accuracy by 0.7 points and AVOID recall by 1.2 points, but actually loses 0.8 points on UNCERTAIN, which foreshadows the trade-off discussed below.

Table 1: Full-supervision ceiling on the 3,637-image test set.

Metric	Hard decoding	Soft decoding
Species top-1 accuracy	0.607	—
Species macro F1	0.484	—
Bucket accuracy	0.824	0.831
Bucket macro F1	0.744	0.750
Recall, SAFE	0.800	0.797
Recall, COOK_FIRST	0.742	0.750
Recall, AVOID	0.895	0.907
Recall, UNCERTAIN	0.535	0.528

5.3. Active Learning Strategies and Learning Curves

We ran the six query strategies for 47 rounds each, going from $N_{\text{init}} = 400$ to $N_{\text{max}} = 5{,}000$ labels in batches of $B = 100$ across five random seeds (a total of 30 trajectories). The learning curves under hard decoding are shown in Figure 4; the soft-decoding version (Figure 5) has the same shape, shifted upward by roughly one point.

Final-round numbers at 5,000 labels are reported in Table 2. Bucket accuracy is tightly clustered: the spread between the best (BALD, 0.789) and the worst (core-set, 0.775) is only 1.4 points, and all six strategies are within 0.6 points of the random baseline. Species accuracy and macro F1 tell the same story.

Table 2: Final-round metrics at 5,000 labels (mean $\pm$ std over five seeds, hard decoding). Best per column in bold.

Strategy	Sp. acc	Sp. F1	Bk. acc	Bk. F1	R_{AVOID}	R_{UNC}
Random	0.513	0.357	0.780	0.682	0.876	0.461
Entropy	0.508	0.372	0.783	0.688	0.875	0.477
Margin	**0.520**	0.382	0.785	**0.693**	0.877	0.487
Core-set	0.501	0.345	0.775	0.676	0.866	0.446
CB-entropy	0.511	**0.387**	0.782	0.690	0.875	**0.489**
BALD	**0.520**	0.363	**0.789**	0.687	**0.896**	0.450
Full-sup ceiling	0.607	0.484	0.824	0.744	0.895	0.535

5.4. Label Efficiency Against the Full-Supervision Ceiling

Table 3 reports the smallest budget at which each strategy first crosses 80%, 90%, and 95% of the full-supervision hard bucket accuracy (0.824). Four observations stand out. First, every strategy reaches the 80% bar within 660 labels, which is about 2% of the training pool. Second, at the 90% bar margin is the most efficient (1,340 labels), nearly halving the requirement of core-set (2,500 labels). Third, at the 95%

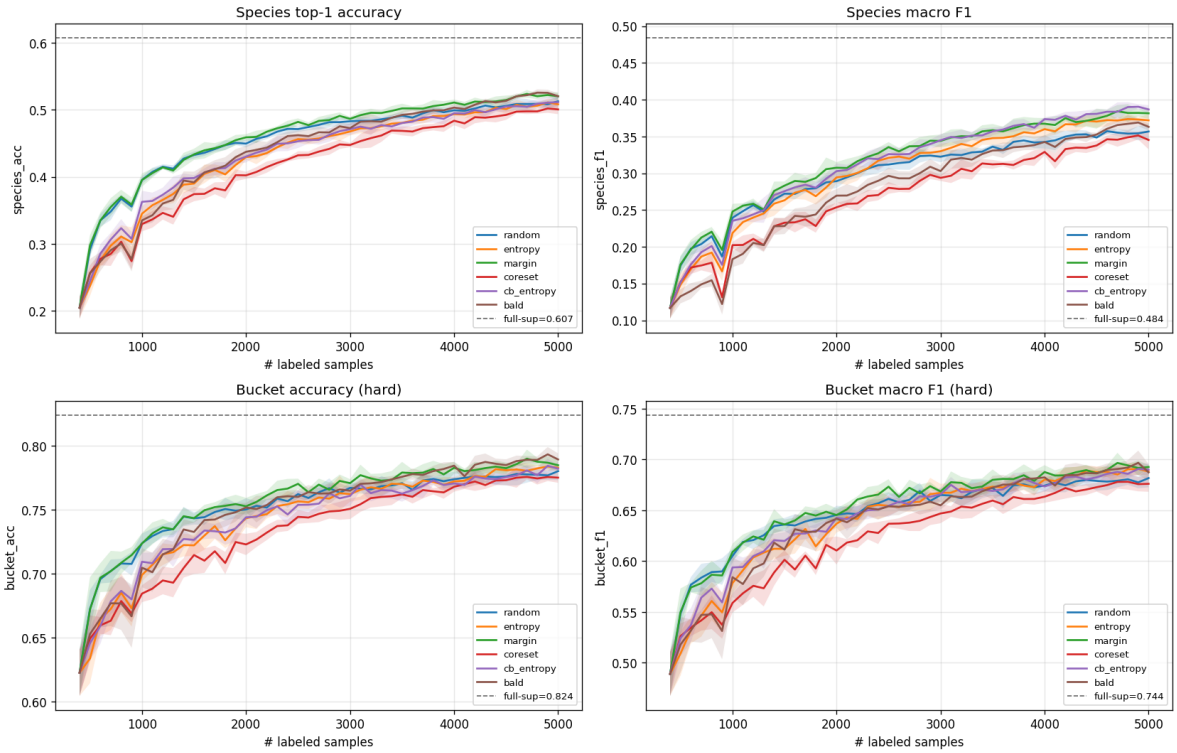


Figure 4: Learning curves for the six AL strategies under hard decoding, mean over five seeds with ± 1 standard deviation as a shaded band. The dashed horizontal line is the full-supervision ceiling. Margin and BALD both reach 95% of the ceiling well before the 5,000-label budget runs out; core-set never closes the gap.

bar the ranking shifts: margin (3,700) and BALD (3,920) both reach the target with all five seeds, while random reaches it with only $2/5$ and core-set never reaches it within the budget. Fourth, taking the soft-decoding view at 95% (target 0.790) makes BALD and margin tie at 3,360 labels, the lowest in the table. The corresponding bar chart is shown in Figure 6.

Table 3: Labels needed to reach τ times the full-supervision ceiling. Cells show the mean budget across seeds, with the success rate (seeds-reached / 5) when not all seeds succeed.

Target (hard, $\tau =$)	Random	Entropy	Margin	Core-set	CB-ent	BALD
80% (0.659)	520	620	**520**	660	660	560
90% (0.741)	1400	1940	**1340**	2500	1980	1660
95% (0.783)	4550 (2/5)	4500	**3700**	N/A	4800 (4/5)	3920
95% soft (0.790)	4600 (4/5)	3920	**3360**	4550 (2/5)	4000	**3360**

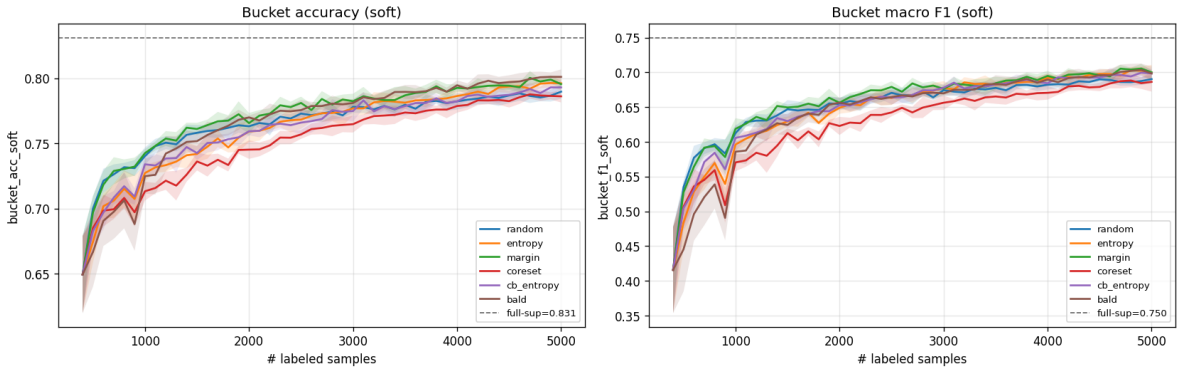


Figure 5: Learning curves under soft decoding. Soft aggregation lifts every strategy by roughly one point on bucket accuracy and bucket macro F1.

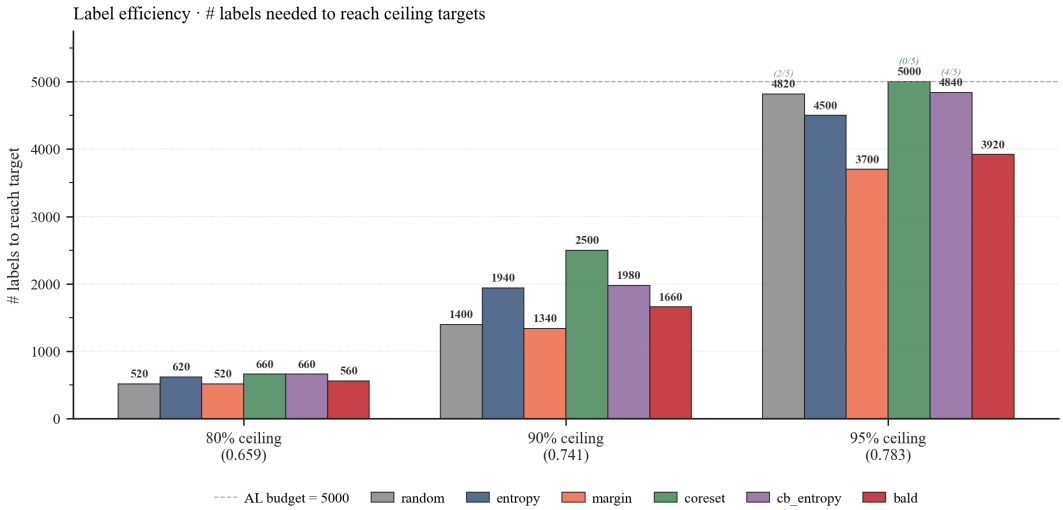


Figure 6: Labels needed to reach 80%, 90%, and 95% of the full-supervision hard bucket accuracy. Margin and BALD are consistently the most label-efficient.

5.5. Per-Bucket Recall and the Safety Trade-Off

Bucket-level accuracy hides where each strategy spends its budget. Figure 7 reports the four per-bucket recall curves. The AVOID curves are tightly clustered around the ceiling, which is the result we want from a safety perspective. The UNCERTAIN curves, in contrast, are noticeably more dispersed and are still climbing at the end of the budget for several strategies, so the choice of strategy matters most precisely for the bucket whose label is "do not eat; flag for research".

The Pareto plot of bucket accuracy against UNCERTAIN recall at the final budget (Figure 8) makes the trade-off explicit. BALD is the top-right corner on bucket accuracy (0.789) but only fifth on UNCERTAIN recall (0.450); class-balanced entropy is the mirror image, top on UNCERTAIN recall (0.489) but third on

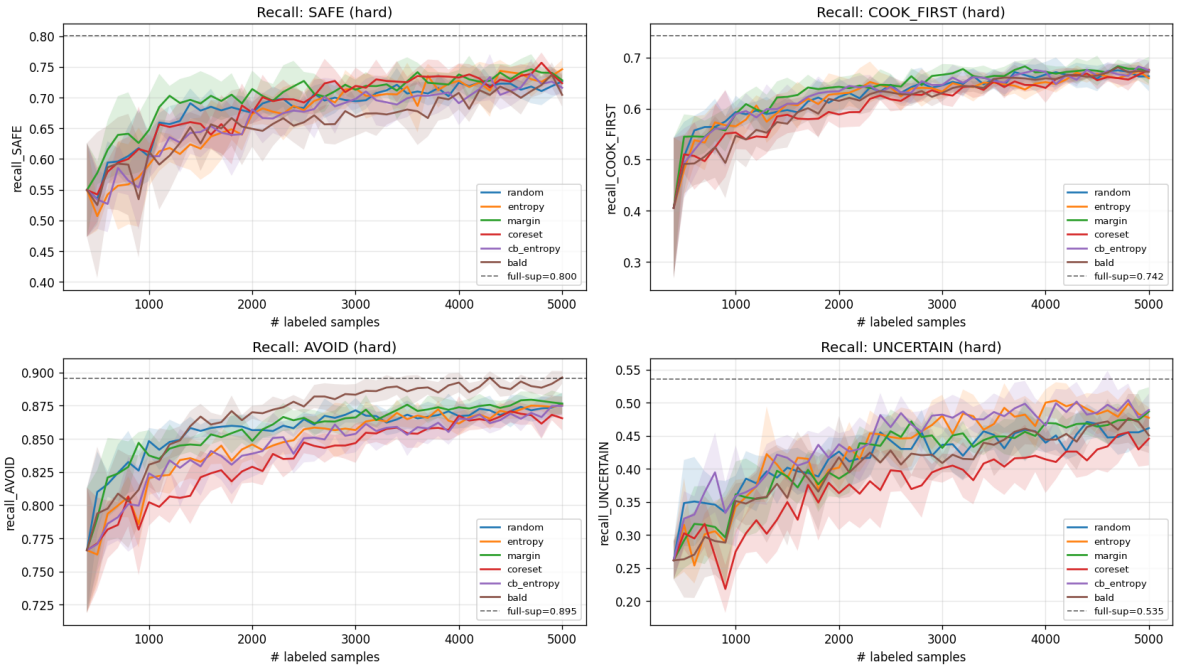


Figure 7: Per-bucket recall as a function of the label budget. AVOID (bottom-left) is recovered uniformly across strategies; UNCERTAIN (bottom-right) is the bucket where strategy choice matters most.

bucket accuracy. Margin sits on the Pareto frontier between the two. The gap to the UNCERTAIN ceiling ranges from 0.046 (CB-entropy) to 0.090 (core-set), so the choice of strategy can move the safety-relevant recall by up to four points.

5.6. Per-Genus Analysis

Table 4 reports per-genus species accuracy at the final budget. The genera order roughly by visual difficulty rather than by strategy: *Boletus* and *Amanita* are the easiest (mean over strategies ≈ 0.65), then *Clitocybe* and *Mycena* (≈ 0.57 and 0.52), and finally *Agaricus* and *Russula* (≈ 0.47 and 0.42). The strategy ranking inside each genus is narrow (≤ 0.06), which means the headline differences between strategies do not come from any one genus.

The most important per-genus result for the safety story concerns *Amanita*, the lethal genus. Although species-level accuracy on *Amanita* only reaches 0.65, the soft-aggregated AVOID recall on *Amanita* is 0.904 and the COOK_FIRST recall is 0.819 (from the genus-bucket breakdown). In other words, even when the model cannot name the exact *Amanita* species, the bucket-level decoder still puts roughly nine out of ten lethal specimens into AVOID. The benefit of the species-then-aggregate decoder in Equation (2) is most

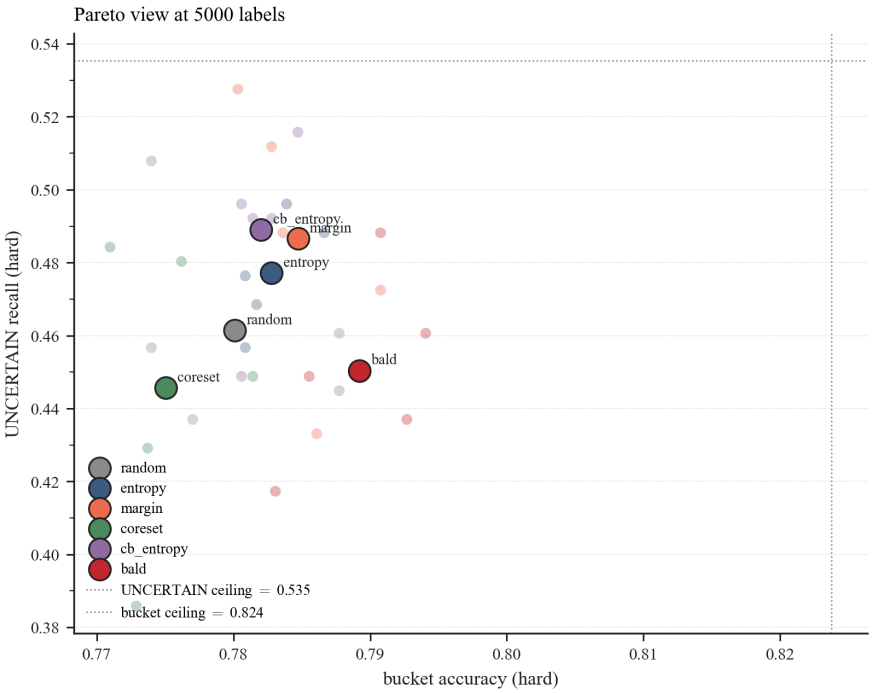


Figure 8: Pareto plot of bucket accuracy versus UNCERTAIN recall at 5,000 labels. BALD optimises overall accuracy, class-balanced entropy optimises UNCERTAIN recall, and margin sits on the frontier.

visible on this genus.

Table 4: Per-genus species accuracy at 5,000 labels (mean over five seeds).

Strategy	Agaricus	Amanita	Boletus	Clitocybe	Mycena	Russula
Random	0.459	**0.667**	0.664	0.557	0.517	0.414
Entropy	0.464	0.647	0.601	0.558	0.508	0.427
Margin	**0.482**	0.651	0.657	0.576	**0.527**	0.425
Core-set	0.466	0.617	**0.665**	0.554	0.507	0.406
CB-entropy	0.471	0.620	0.625	0.542	0.526	**0.431**
BALD	0.446	**0.674**	0.656	**0.601**	0.508	0.437

6. Discussion

6.1. Soft Decoding Forgives Within-Bucket Confusion

The systematic gain from hard to soft decoding (+0.7 points on bucket accuracy and a much larger swing on the per-genus AVOID recall for *Amanita*) confirms our motivating intuition: training on 179 species and aggregating only at the end is more robust than collapsing the labels first. The model is allowed to be confused between two *Boletus* edibles or two *Amanita* toxics, as long as both candidates fall inside the same

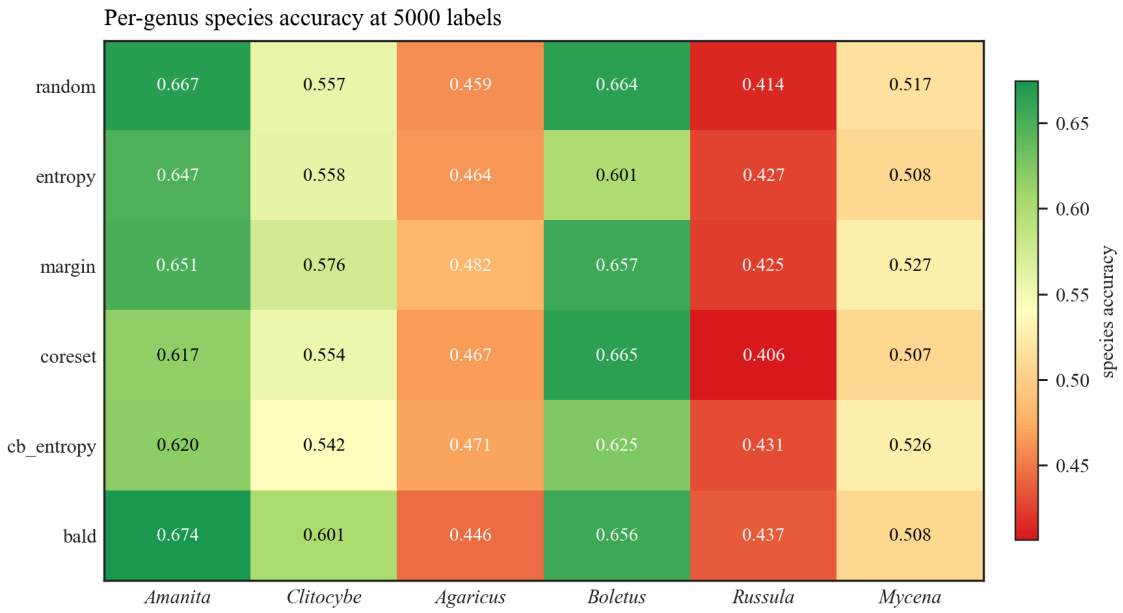


Figure 9: Per-genus, per-bucket recall heatmap. *Amanita*'s AVOID row (0.89) is the safety-critical cell and is reached even though its species accuracy is only 0.65.

bucket. The one bucket that does not benefit from soft decoding is UNCERTAIN, which is consistent with its within-bucket members being scattered across genera (mostly *Russula* plus a few *Agaricus* and *Clitocybe*); summing probabilities across visually unrelated species does not help it.

6.2. No Single AL Strategy is Best on Every Metric

The headline finding from Section 5.4 is that the six strategies are very close on bucket accuracy (1.4-point spread) yet noticeably separated on UNCERTAIN recall (4.3-point spread) and on label efficiency. This is the opposite of the usual AL story, in which uncertainty methods clearly beat random on a single overall metric. With a strong, fungi-specialised backbone the marginal value of any one strategy is small, and the practical question becomes *which trade-off to take*, not *which strategy is best*. The Pareto plot in Figure 8 should therefore be read as the main takeaway, not the final-round accuracy column.

6.3. Why BALD Wins Overall but Loses on UNCERTAIN

BALD targets epistemic uncertainty, that is, examples where the MC-dropout posterior disagrees with itself. In our setting that disagreement concentrates inside large, visually crowded clusters (the AVOID *Mycena* and *Russula* species), which is why BALD reaches 0.896 AVOID recall, matching the full-supervision ceiling, with only 15% of the labels. The same mechanism hurts it on UNCERTAIN: those species have only 2,553

images total (7% of the pool), and the posterior agrees that they are unfamiliar rather than disagreeing, so they are passed over for examples from the larger AVOID pool. Class-balanced entropy goes the other way by construction, paying for its UNCERTAIN recall with a small drop in overall accuracy. Margin is the only strategy that is on the frontier on both axes and is the most label-efficient at the 90% and 95% targets, which makes it our recommended default if only one strategy is to be deployed.

6.4. Where the Headroom Actually Is

The residual gap to the full-supervision ceiling at 5,000 labels is about 4 points of bucket accuracy, but the gap from full-supervision to a hypothetical perfect classifier is 17 points. In other words, even an unlimited annotation budget on the current pipeline leaves more headroom than active learning could ever recover. The bottleneck is the frozen ViT-B/16-224 representation: at 0.626 logistic-regression species accuracy, it is more than 0.1 below the full-finetune number reported by Picek et al. [1] and well below the larger ViT-L/16-384 SOTA. Future gains will come from training the backbone (or at least the last few blocks) inside the AL loop, not from squeezing additional points out of the strategy. The small spread between strategies in Table 2 is direct evidence of this: when the representation is the bottleneck, query selection has limited leverage.

6.5. Implications for Safety-Critical Deployment

Three results map directly onto a foraging-app deployment. First, AVOID recall is essentially solved at 5,000 labels for every strategy: the classifier catches roughly nine out of ten toxic specimens, and that number does not depend on which AL method was used to build the labeled set. Second, the *Amanita*-specific AVOID recall of 0.90 under soft decoding implies that, at this label budget, a wild-mushroom user who is shown the predicted bucket and is told to defer when UNCERTAIN comes up would be protected from the deadliest mistakes nine times out of ten. Third, UNCERTAIN recall is the metric that genuinely depends on strategy choice; if the deployment surfaces an explicit "defer to an expert" channel rather than just a top-class label, picking class-balanced entropy or margin over BALD changes how often that channel fires by several points.

6.6. Limitations

We use a frozen backbone that was itself fine-tuned on the same DF20-Mini training pool we sample from, which makes our embeddings an upper bound on what a true cold-start AL pipeline would see. We also keep the head small and retrain it from scratch each round, both for runtime reasons and because warm-starting biased the BALD scores in early experiments; we did not study fine-tuning of the backbone inside the AL loop. The edibility labels come from a single LLM call with no expert review, and the AVOID bucket lumps two deadly species together with 89 merely poisonous or inedible species. Finally, all experiments are on Danish fungi; transfer to a new region would require both a new embedding pass and at least a partial re-labeling of the edibility hierarchy.

References

1. L. Picek, M. Šulc, J. Matas, T. S. Jeppesen, J. Heilmann-Clausen, T. Læssøe, and T. Frøslev, "Danish Fungi 2020 – Not Just Another Image Recognition Dataset," in *Proc. IEEE/CVF Winter Conf. on Applications of Computer Vision (WACV)*, 2022, pp. 1525–1535.
2. L. Picek, M. Šulc, J. Matas, J. Heilmann-Clausen, T. S. Jeppesen, and E. Lind, "Automatic Fungi Recognition: Deep Learning Meets Mycology," *Sensors*, vol. 22, no. 2, p. 633, 2022.
3. B. Settles, "Active Learning Literature Survey," *Computer Sciences Tech. Report 1648*, University of Wisconsin–Madison, 2009.
4. O. Sener and S. Savarese, "Active Learning for Convolutional Neural Networks: A Core-Set Approach," in *Proc. Int. Conf. on Learning Representations (ICLR)*, 2018.
5. Y. Gal, R. Islam, and Z. Ghahramani, "Deep Bayesian Active Learning with Image Data," in *Proc. Int. Conf. on Machine Learning (ICML)*, 2017, pp. 1183–1192.
6. Y. Gal and Z. Ghahramani, "Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning," in *Proc. Int. Conf. on Machine Learning (ICML)*, 2016, pp. 1050–1059.
7. U. Aggarwal, A. Popescu, and C. Hudelot, "Active Learning for Imbalanced Datasets," in *Proc. IEEE/CVF Winter Conf. on Applications of Computer Vision (WACV)*, 2020, pp. 1428–1437.